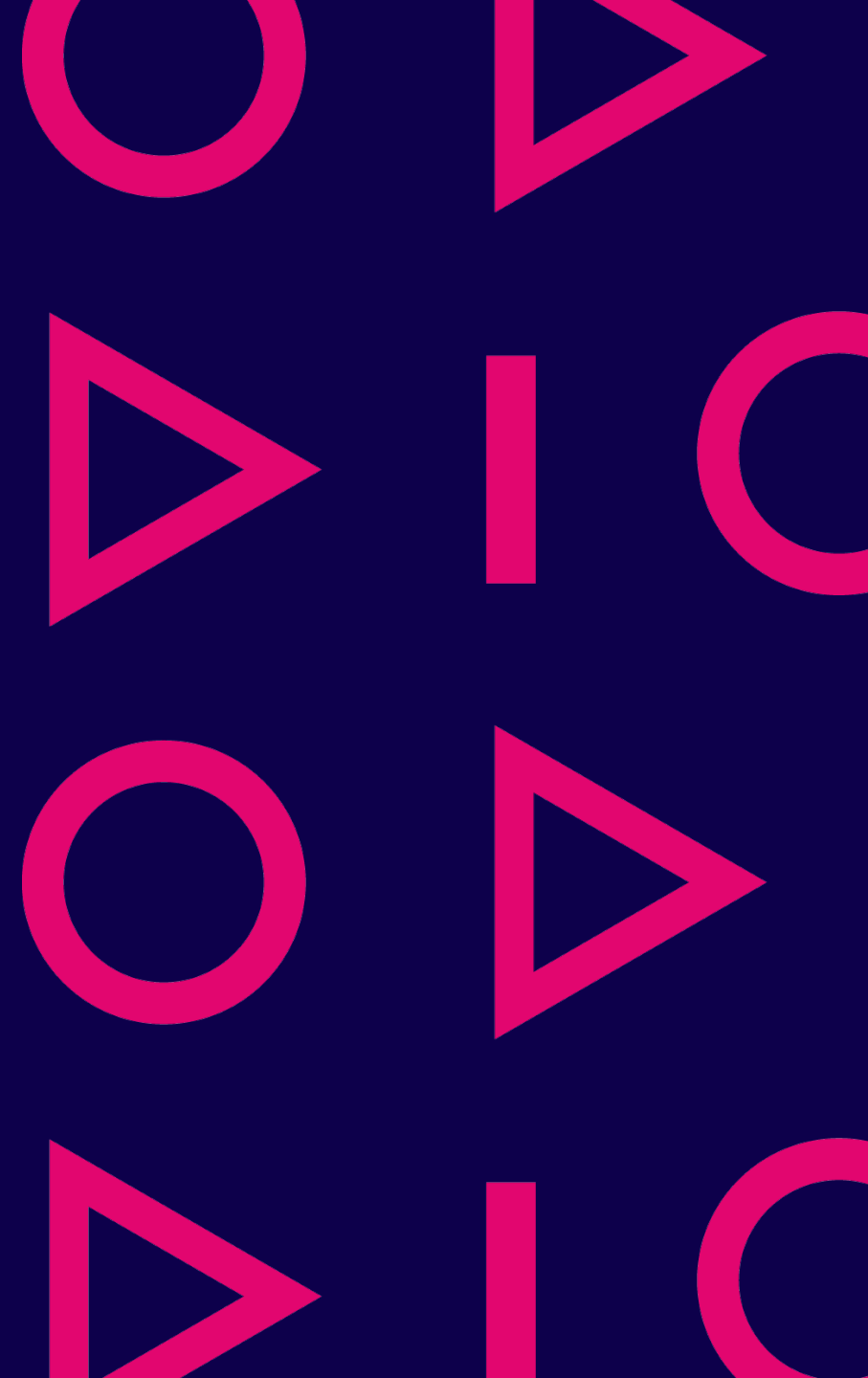
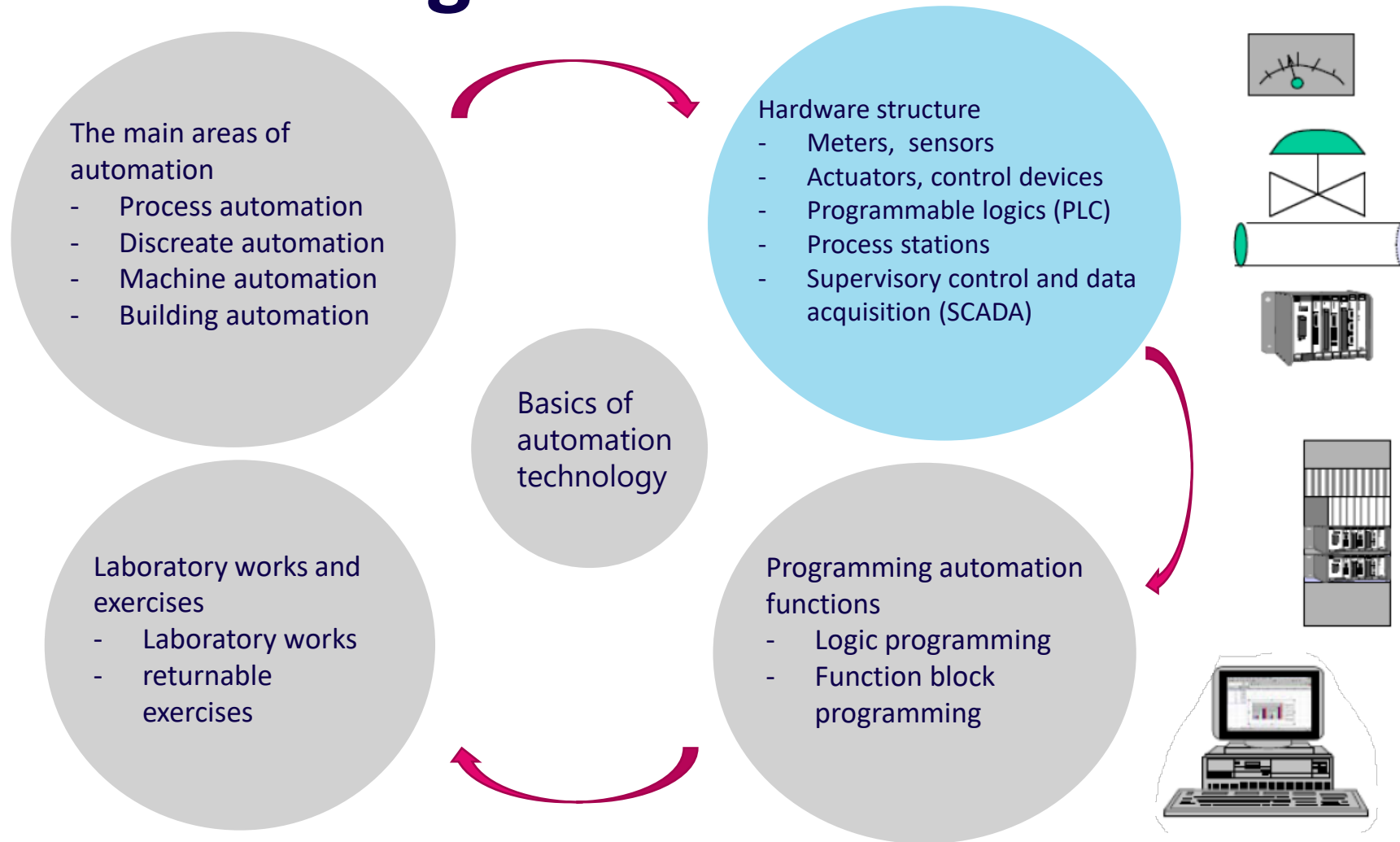


Basics of automation technology

Samppa Alanen

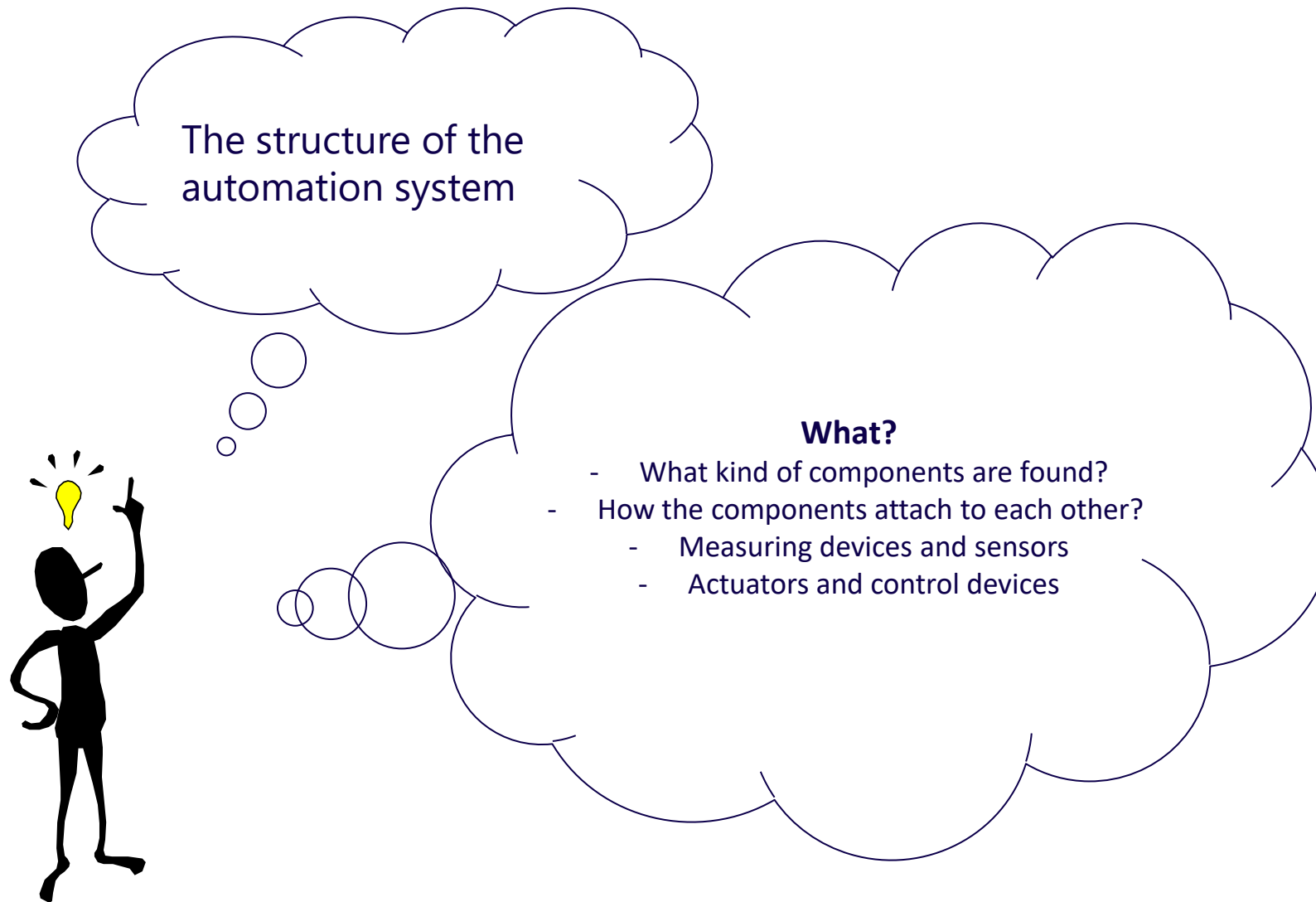


Course flow diagram



Content of the 2nd lesson

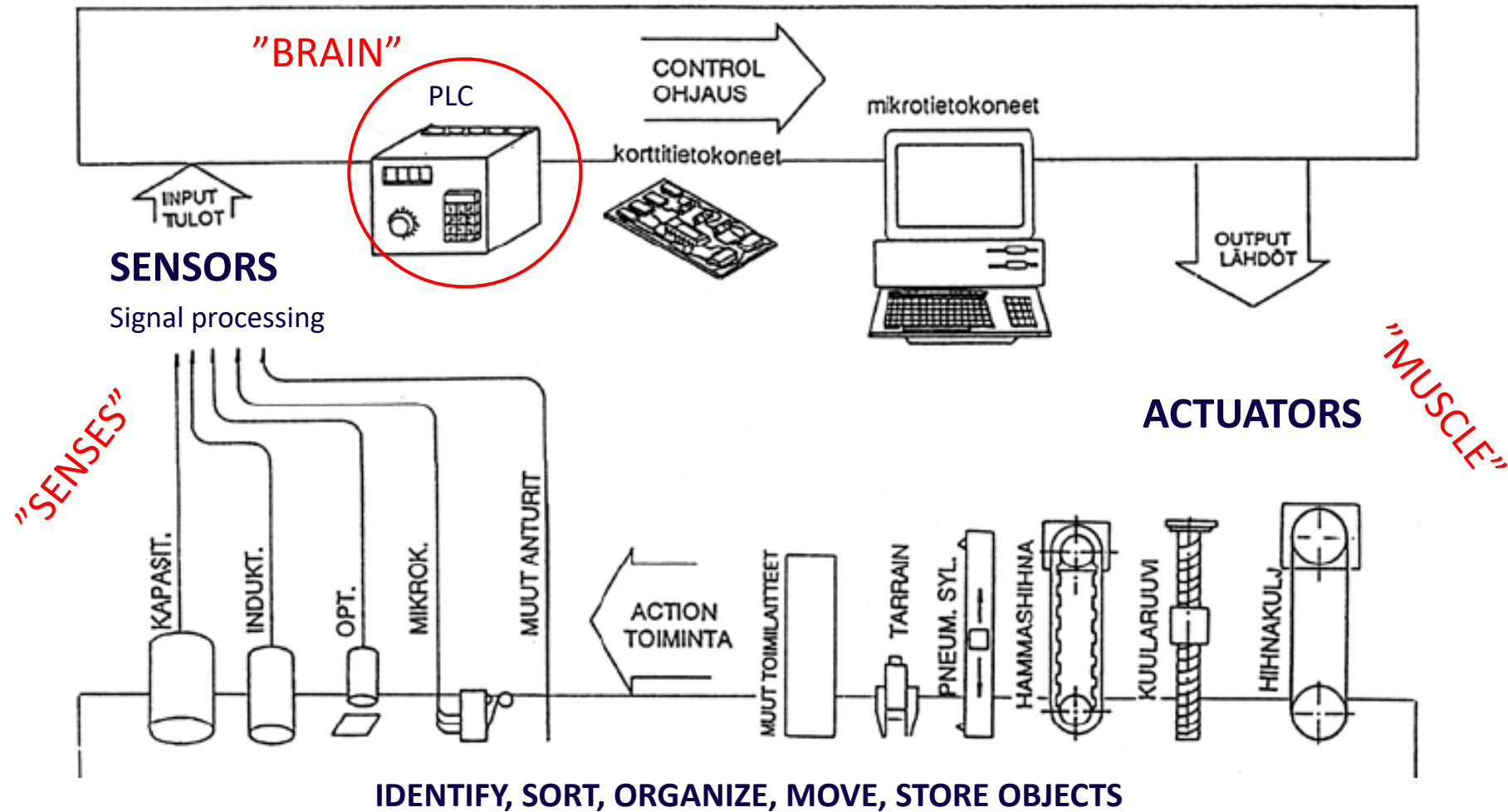
- General principles of the automation/control system, hierarchical levels. DCS vs. PLC
- Examples: Visual Components and Automation Studio
- Overview of indicators/sensors and actuators. (Analog measurements will be discussed later in the context of analog signals)
- PLC - Programmable Logic Controller
- I/O, binary inputs
- Automation Studio example: converting an electropneumatic coupling to a logic-controlled one



Reference: Seppo Rantapuska

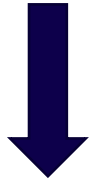
Automation equipment

CONTROL SYSTEM

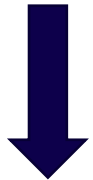


The general principle of automation

Measurement/sensors/indicators



Programmable logic or process station



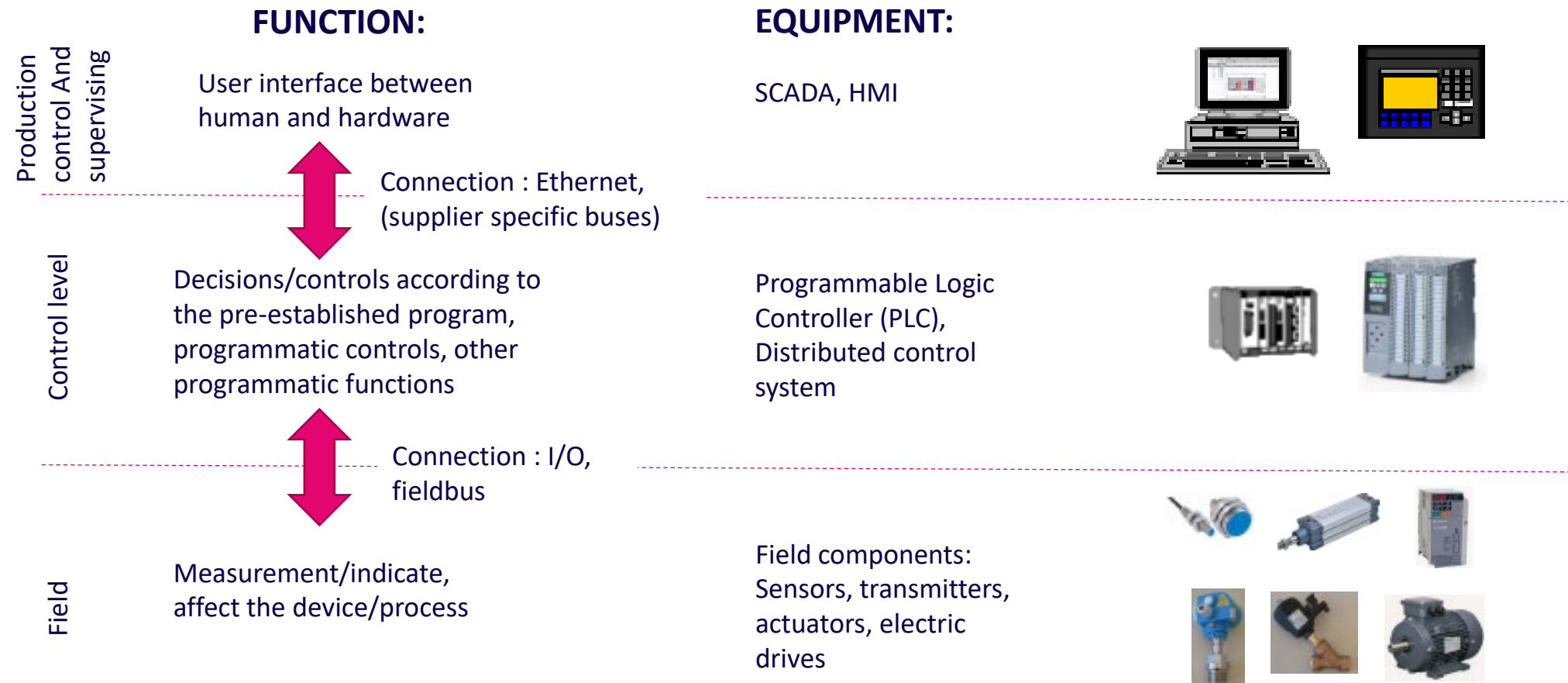
Actuators

"Senses" by which automation can observe the state of a process or device. For example, proximity switches.

"Brain". Make decisions/controls according to the program based on the data obtained from the measurements. E.g. Sensor detects tank is full → stop filling pump

"Muscles", controlled by programmable logic, can affect the state of the device / process. For example, a pneumatic cylinder, an electric motor, etc.

The three hierarchical levels of automation



Distributed control system DCS

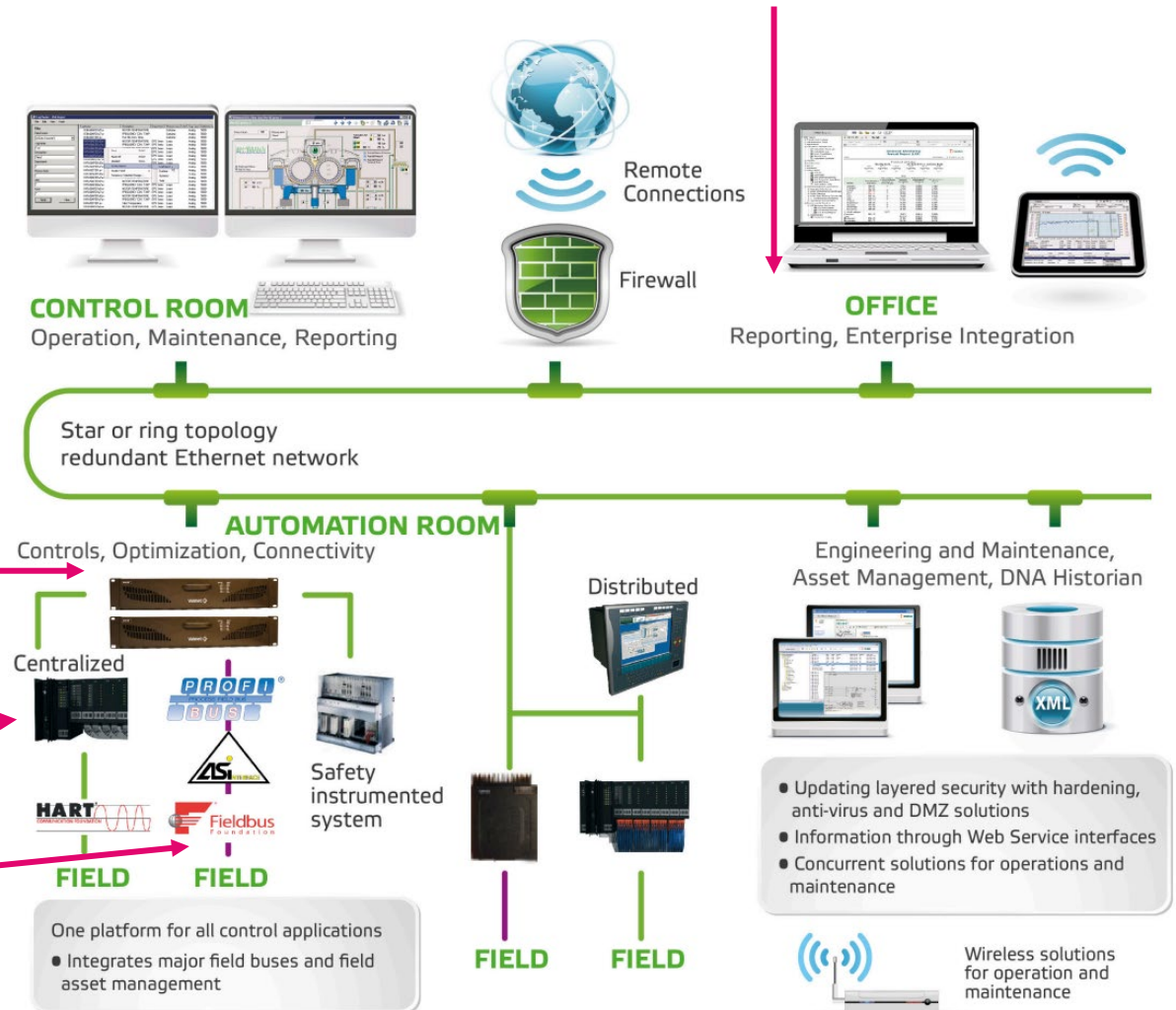
- Typically for managing large process entities (I/O scope up to tens of thousands of measurements and controls – > power plant, paper machine, pulp mill, etc.)
- Includes control room level (+ history data collection, design server, etc.) and automation level (process stations and IO) from the same manufacturer/supplier
- Cycle time typically 100ms – 1000ms
- In this course, the DCS system will not be discussed further

Plants may have systems above the management level (MES) for which orders are given to the automation system

PCS is a process station that runs an automation application loaded on it (a bit like a PLC's CPU unit)

IO rack (contains IO modules/cards to which signals from "traditional" field devices can be connected)

Devices connected directly to the fieldbus



Programmable logic controller PLC

- Small and medium-sized projects (e.g. < 300 I/O → Individual machines/equipment, conveyor systems, smaller processes, etc.)
- Freely programmable device for controlling process/machine/equipment actuators based on measurements connected to the equipments
- Made for industrial use → Good reliability, resilience, etc.
- Long life cycle. Up to 20-30 years of replacement and spare parts availability from large manufacturers
- Standard connections enable changes to the control system when needed so that the entire field device level does not have to be rebuilt
- Standard programming environments and similar operating principles make it possible to learn a new logic character relatively quickly
- Fast cycle times (milliseconds)
- Modular (expandable with modules) or compact (not expandable)

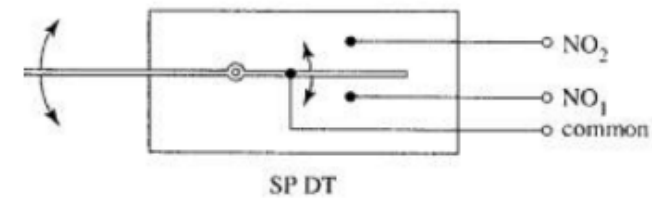
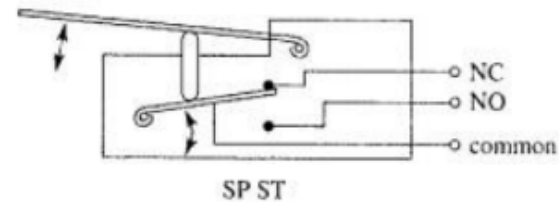
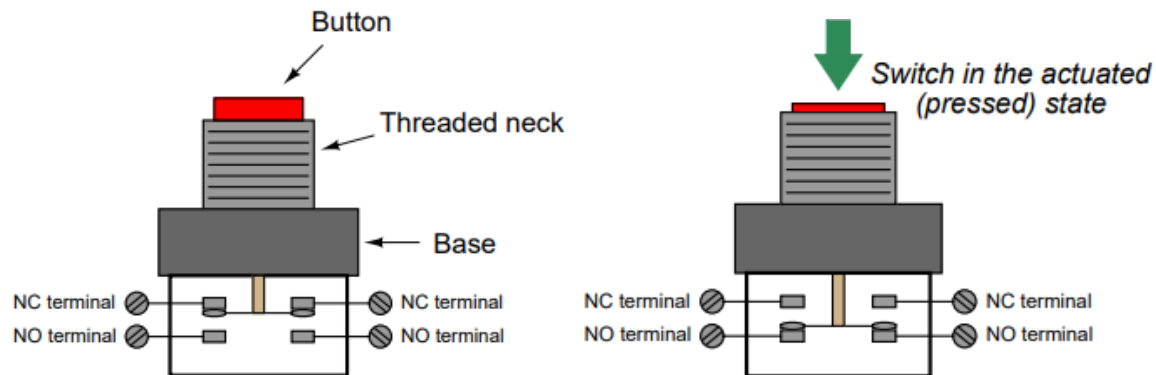
In the exercises of this course, in labs, etc. focus on entities that can be managed with programmable logic.



Visual Components examples

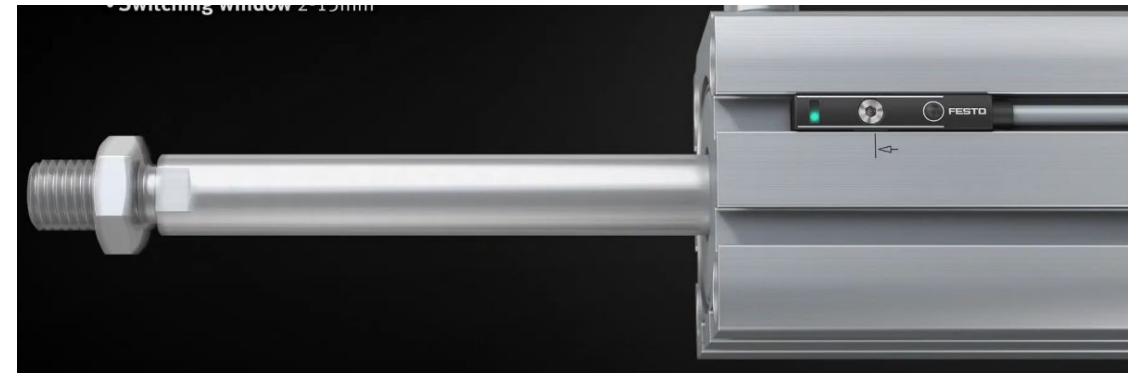
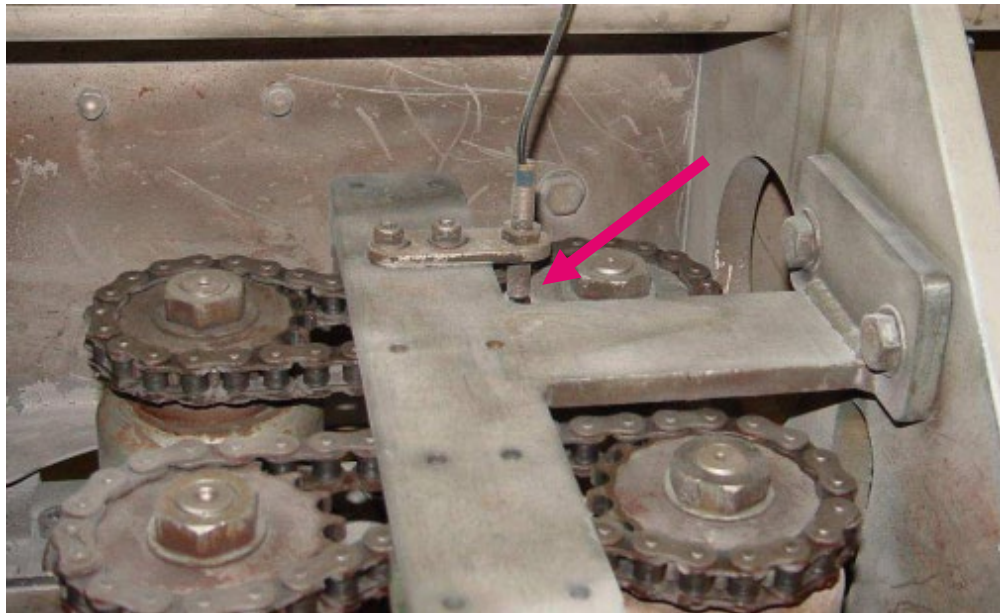
Detectors and sensors

- Mechanical switches
 - Push buttons, door switches
- Normally-open, Normally-closed and Change-over contacts



Detectors and sensors

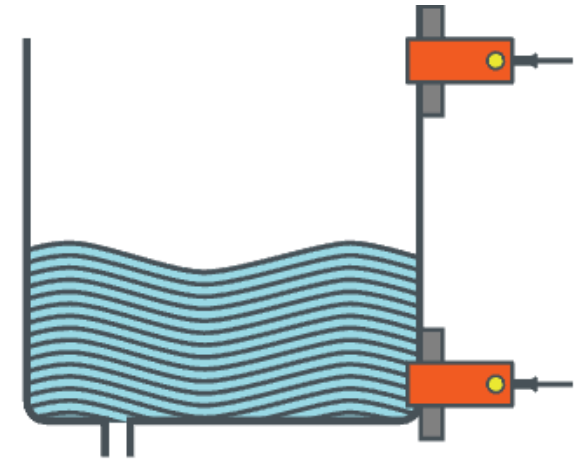
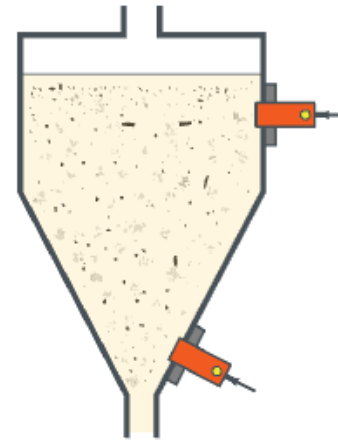
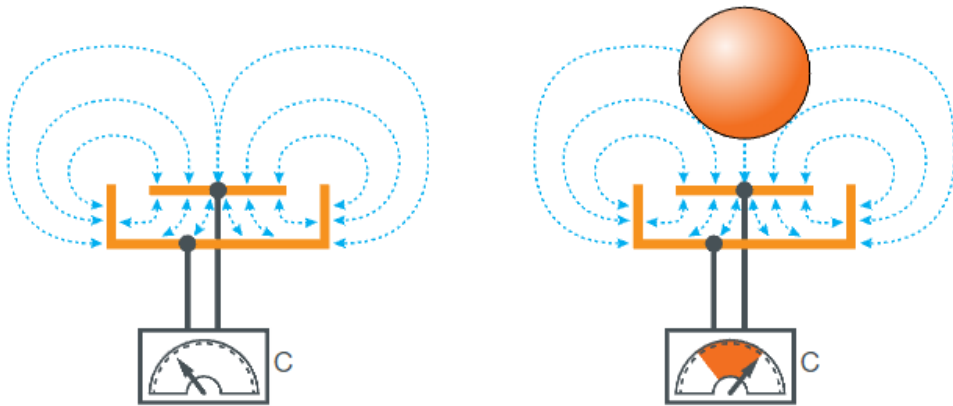
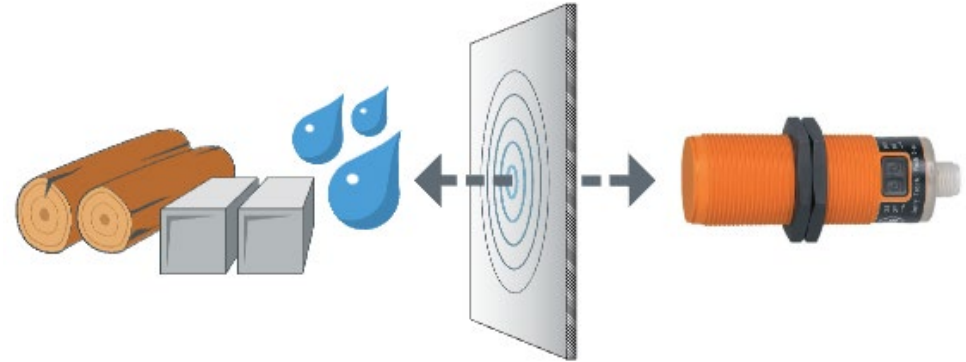
- Inductive proximity sensor for ferromagnetic materials
- Non-contact, detection distance mm to cm
- Cylinder limit switches, rotation detection from the sprocket (is the machine running), ABS brake
- High-speed switching frequency



<https://www.ifm.com/us/en/us/overview/inductive/technology-overview/tech-overview>

Detectors and sensors

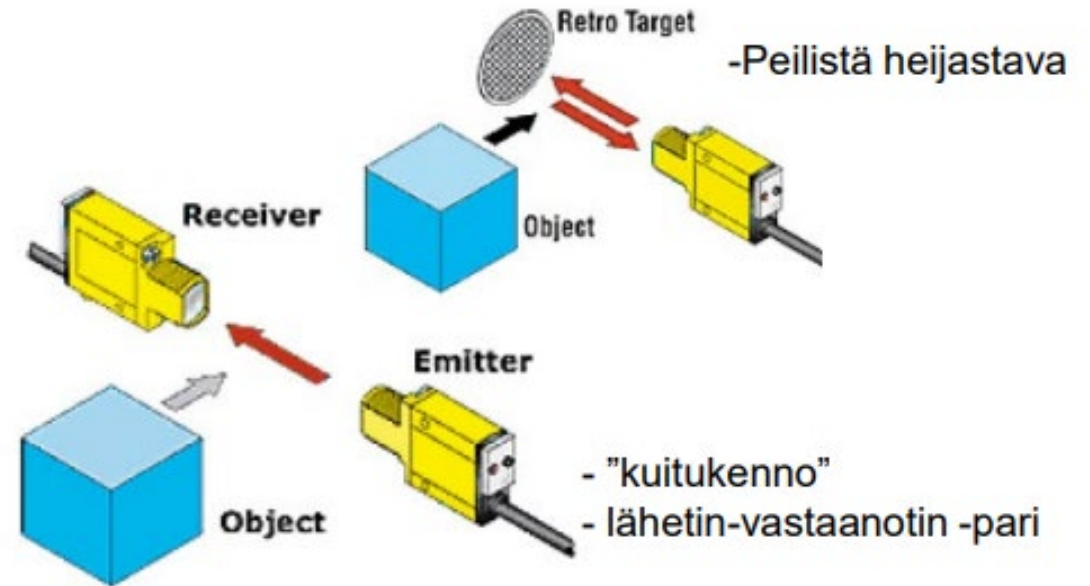
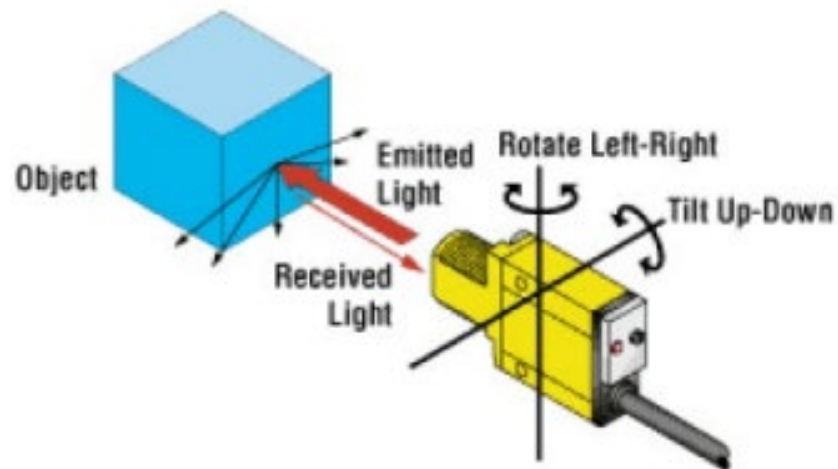
- Capacitive proximity sensor
- Contact or non-contact
- Able to measure through non-metallic materials
- Adjustable sensitivity
- Lower switching frequency compared to inductive



<https://www.ifm.com/us/en/us/overview/capacitive/technology-overview/technology-overview>

Detectors and sensors

- Optical sensor
- Reflection, transimission
- <https://www.ifm.com/us/en/us/overview/photoelectric/technology-overview/tech-overview>



Cylinders and directional valves

- The principle of single-acting and double-acting cylinder
- The principle of directional valves

Review of pneumatic valves:

<https://www.omega.co.uk/auto/pdf/SimpValvesGuide.pdf>

Lessons In Industrial Instrumentation, ibiblio.org, Chapter 10.

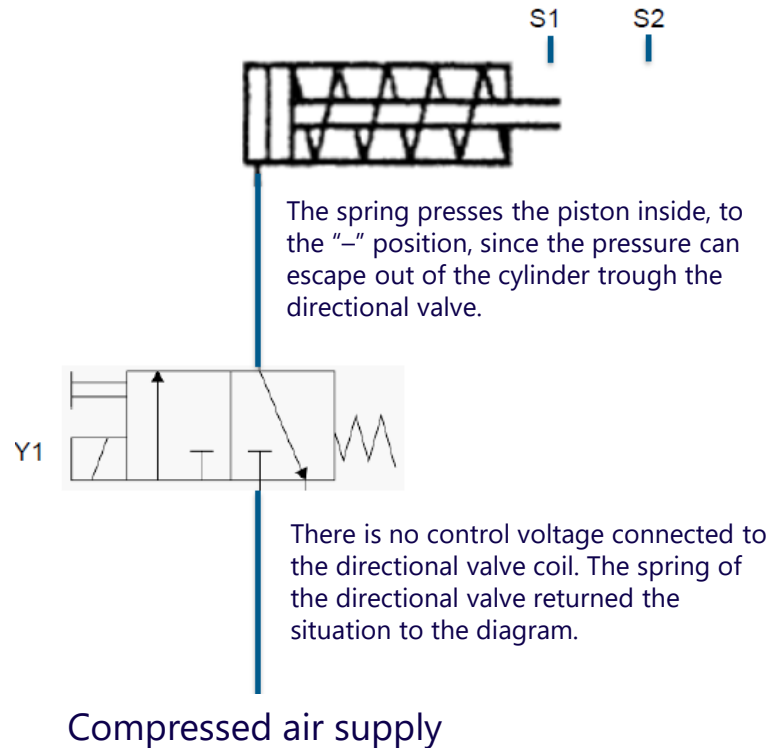
Explore the principle of operation of the pneumatic cylinder and the directional valve that controls it, watch the following videos:

<https://youtu.be/Jfdmrm4A99s>

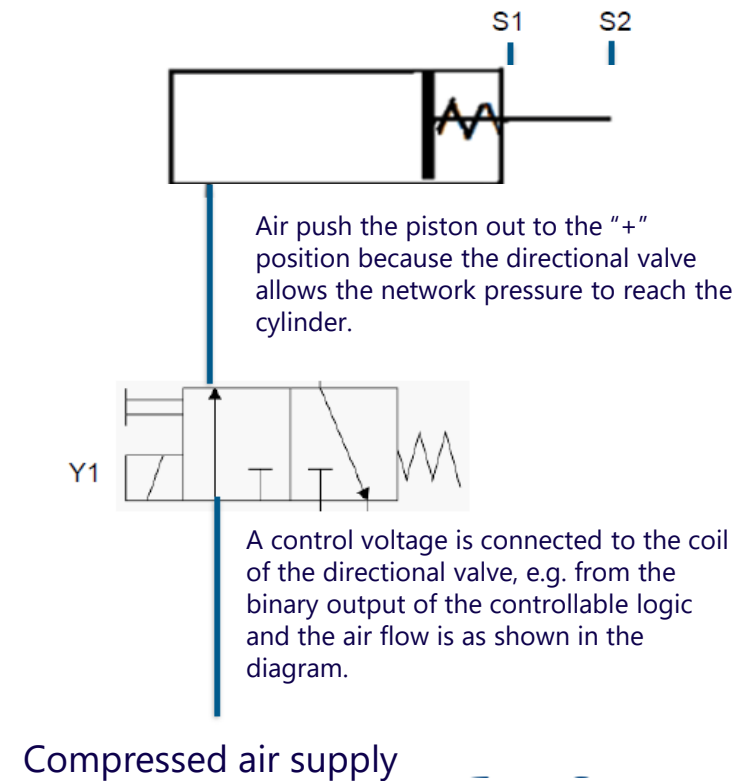
<https://youtu.be/YppUetnI-6M>

<https://youtu.be/R-OBtVCPjMc>

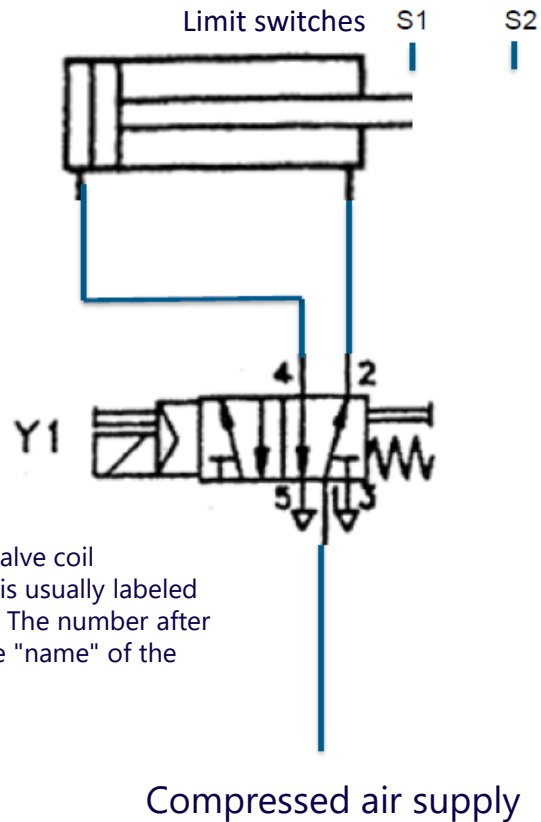
Control of a single-acting cylinder with a 3/2 directional valve



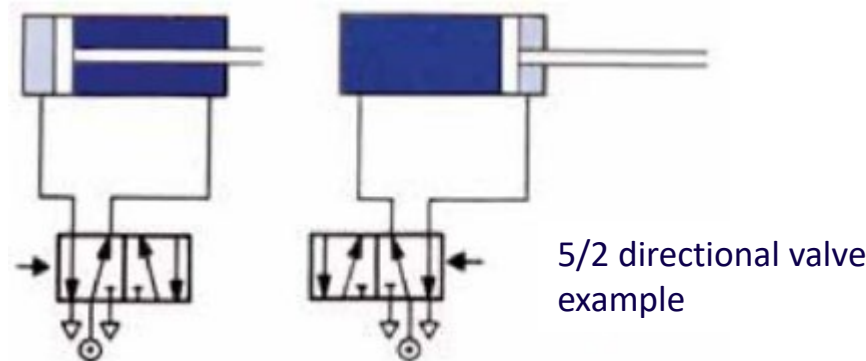
S1 ja S2 ovat rajakytkimet, joilla tunnistetaan onko mäntä ulkona vai sisällä.



Control of a double-acting cylinder with a 5/2 directional valve



The directional valve coil (electromagnet) is usually labeled by the symbol Y. The number after the symbol is the "name" of the spool.



Directional valves



2-position, 2-way, 2 ported



2-position, 3-way, 3 ported



2-position, 4-way, 4 ported



2-position, 4-way, 5 ported



3-position, 4-way, 4 ported Closed Center



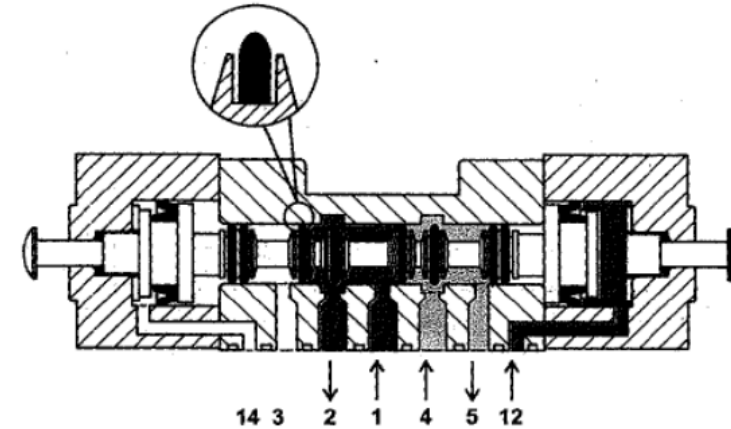
3-position, 4-way, 5 ported Closed Center



3-position, 4-way, 5 ported Pressure Center



3-position, 4-way, 5 ported Open Center



1 Air supply

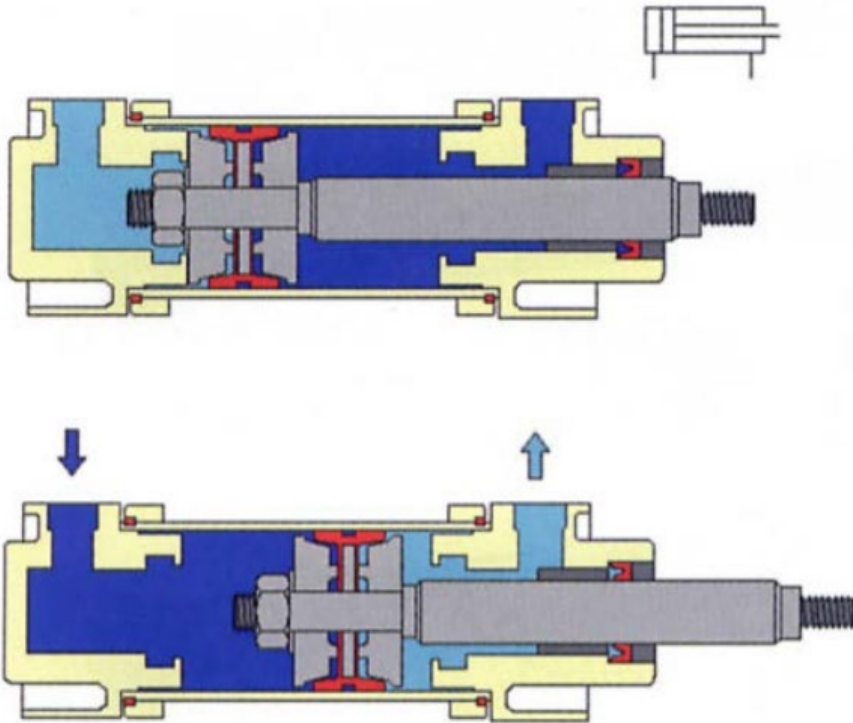
2 and 4 actuator connection

3 and 5 exhaust

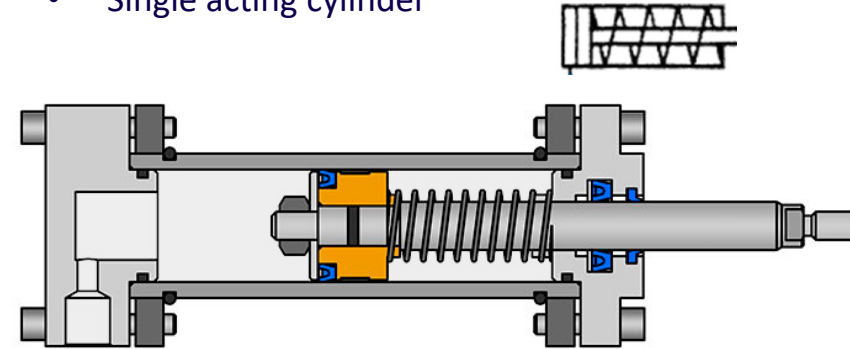
12 and 14 control (air)

Structure of cylinders

- Double acting cylinder



- Single acting cylinder



Connections to the field level

- The automation level(programmable logics and process stations) is attached to the field level (measuring and actuators) either:
 - Directly on fieldbuses (intelligent field devices connected to serial communication)
 - I/O modules connected to the fieldbus (“distributed I/O”)
 - With I/O connected directly to the controller (IO = Inputs/Outputs, so-called traditional "parallel cabling", with its own wires/pairs of wires for each signal/device)
- The most common types of IO are (**IMPORTANT TO KNOW**):
 - Binary Input (**BI** or **DI**) → To connect on/off signals; proximity switch, surface switch, push-button or other "switch information"
 - Binary Output (**BO** or **DO**) → on/off controls; engine start/stop, shut-off valve open/closed, light on/off, etc.
 - Analog Input (**AI**) → continuous measurements; Temperature measurement, pressure measurement 0-6 bar, level measurement 0-100%. (0...10V, 4...20mA)
 - Analog Output (**AO**) → Continuous control; engine speed control, control valve control 0-100%, etc. (0...10V, 4...20mA)

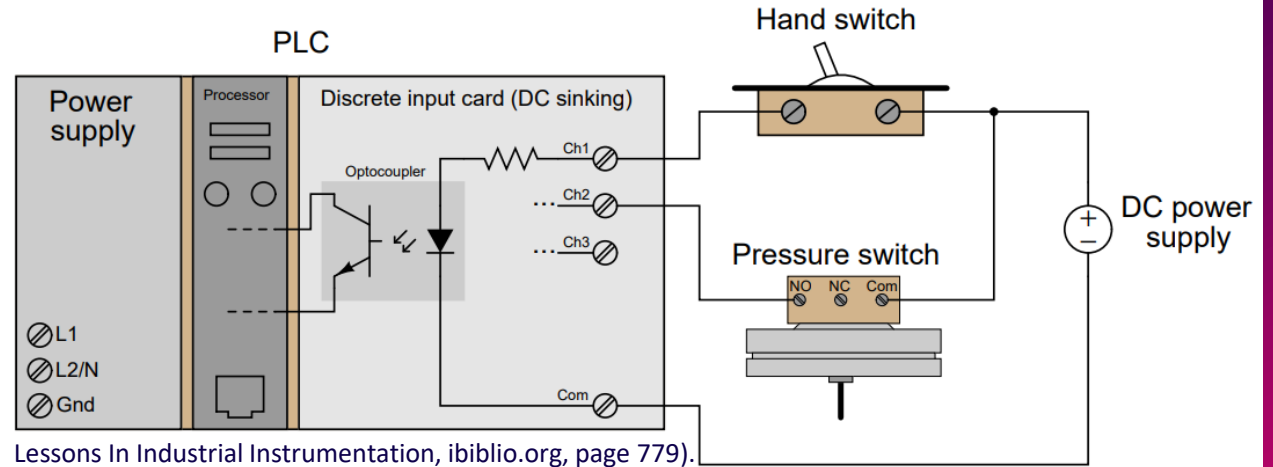
PLC binary (discrete) input channel

When the contact of the switch (Hand Switch) closes, the supply voltage can act on the channel terminal screw 1 (Ch1) of the binary input card, so that current passes through the LED of the optocoupler belonging to the internal interface electronics of that channel to the common connector on the input card and from there back to the voltage source (closed circuit).

When the current passes through the opto-isolator led as mentioned above, the phototransistor on its "secondary" side begins to conduct (under the influence of light), at which point the one-bit sized input address is placed in state 1 (true) in the memory space of the logic CPU. When the power goes out the address changes back to 0 (false).

Based on the status of the input address, the program performs the actions that the closing switch should occur (e.g. turn off the lights).

The status of the pressure switch can be read from the input channel 2 (Ch2, the same principle, as in channel 1).

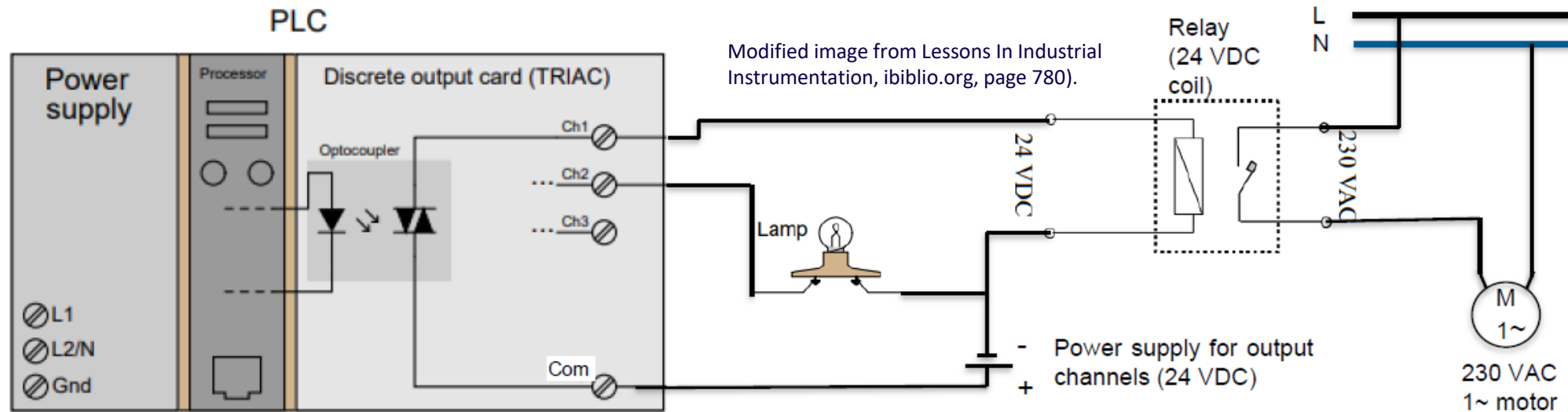


Energizing an input channel lights the LED inside the optocoupler, turning on the phototransistor, sending a "high" signal to the PLC's microprocessor, setting (1) that bit in the PLC's input register.



Siemens S7-300 logic

PLC binary (discrete) output channel

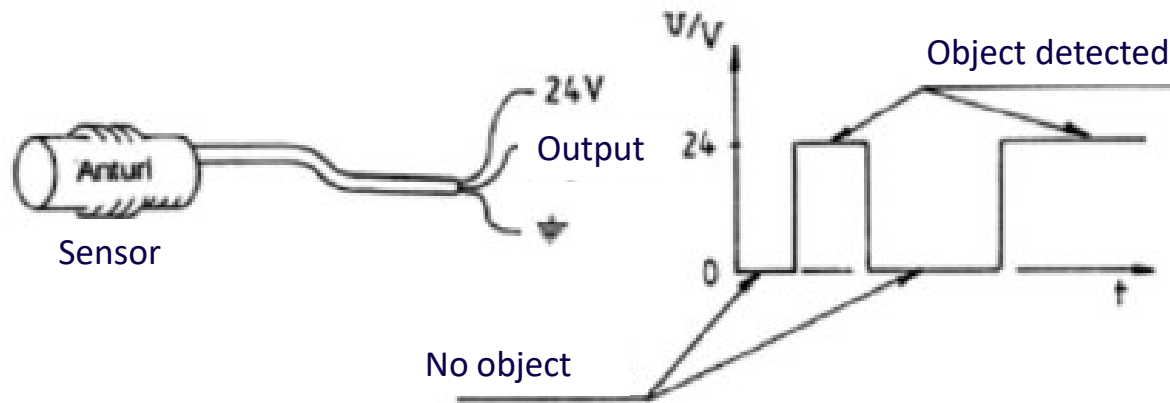


When the one-bit-sized output address of to the logic's binary output channel 1 (Ch1) is set to 1 (true) in the program, the CPU directs the output channel on. A current start to flow through the LED and emitted light falls on a phototransistor cause a current flow through the phototransistor. In this case, the terminal screw of channel Ch1 becomes energized (an external 24VDC source supplying the output card) and current can pass through the coil of the relay connected to that channel. As a result, the relay pulls: its contact closes and the 1-phase engine on the 230 VAC main circuit starts. Accordingly, by controlling channel two, a (24VDC) lamp connected to it can be turned on or off.

There could be a mechanical relay in place of the optocoupler ("relay output"). In any case, both input and output modules are available in different types of electrical operation (i.e. the electrotechnical implementation of the channels varies)! These will be discussed in more detail in future courses.

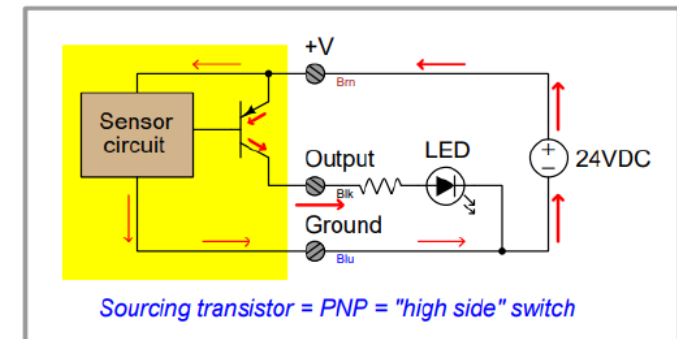
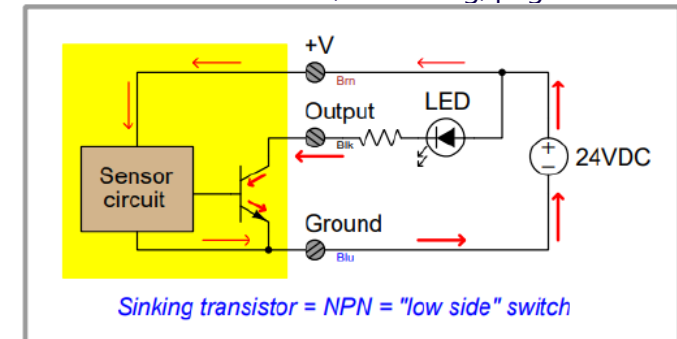
Sensor coupling (proximity switch)

- The figure shows the principle of connecting a three-wire proximity switch. This may be, for example, the case of: inductive or capacitive proximity switch that needs an operating voltage (+ and -, 24VDC). The third wire is the signal from the sensor. Depending on whether it is a PNP or NPN type sensor, the output signal switches to the level of the sensor supply voltage + or – when the sensor reacts.

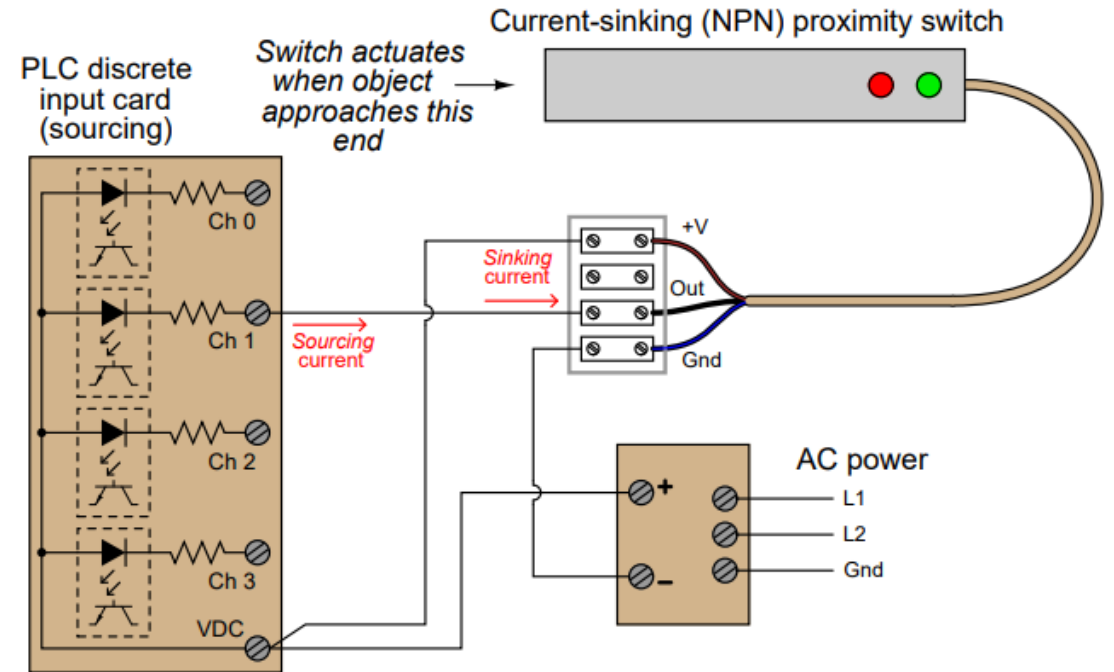
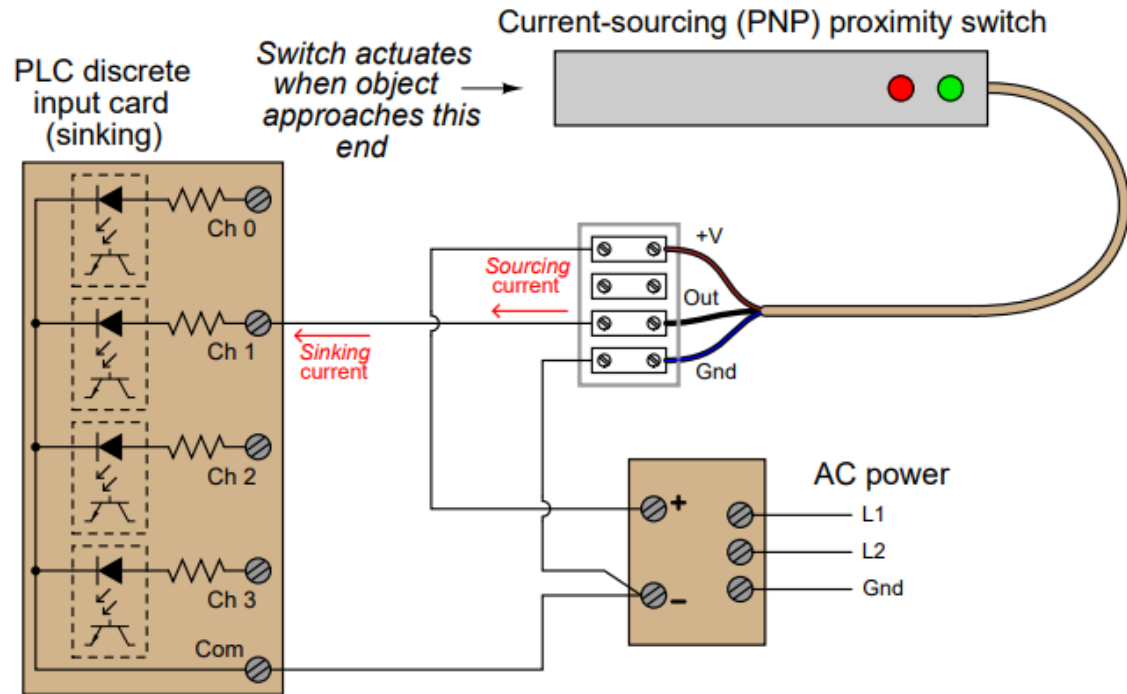


- The output signal is connected to the binary input channel in the I/O of the logic/process station. In this case, the minus (-) of the supply voltage to the sensor (24V + and -) must be in the same potential with the common terminal of the input module. Otherwise, there will be no closed circuit! The example illustrates the signal given by a PNP-type sensor.

- The fundamental difference between the PNP and NPN connection (LED is in the example as a load, that is, the light is on when the sensor reacts. More information: Lessons In Industrial Instrumentation, ibiblio.org, pages 666 and 667):



Sensor coupling (proximity switch)

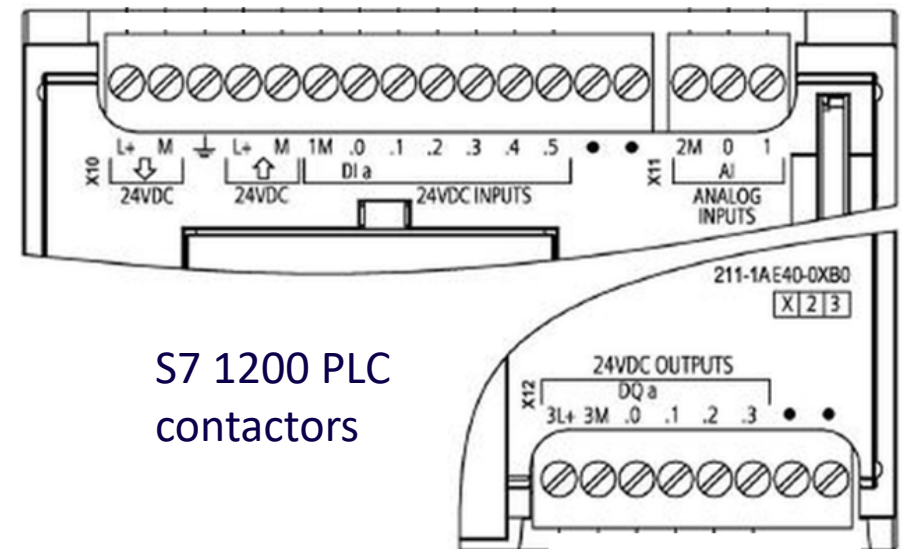
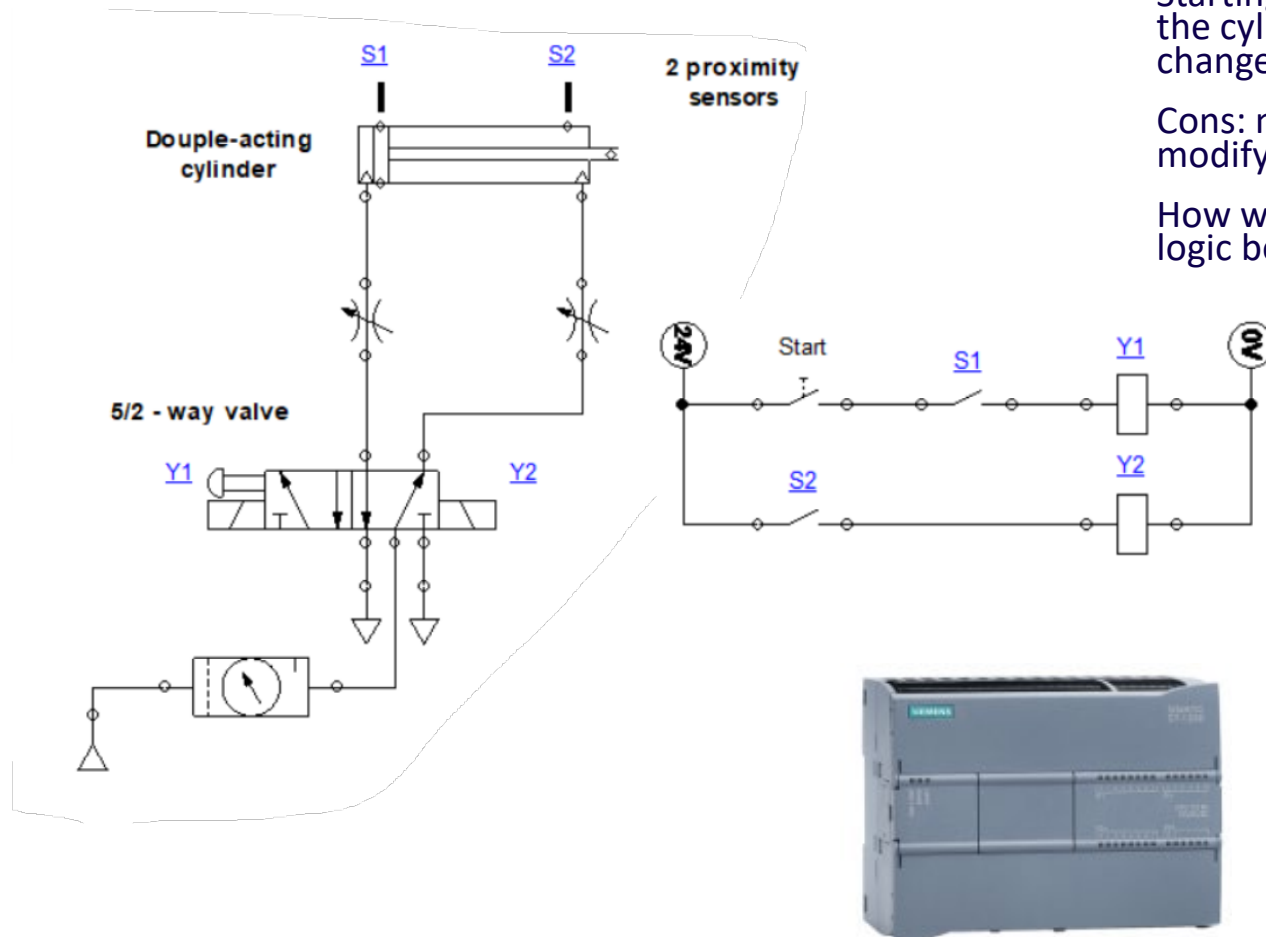


Example: changing the cylinder control to Siemens programmable logic (AutomationStudio example) page 1/2

Starting point: the so-called. wired logic (fixed coupling) controls the cylinder out and in (continuous movement, limit switch changes direction) when the Start contact is closed.

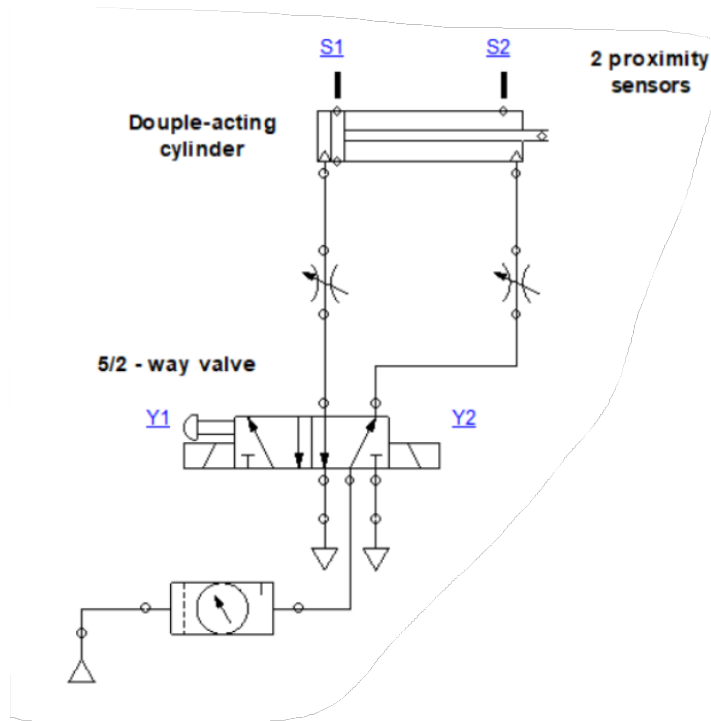
Cons: no timers/delays, no calculation of work cycles, difficult to modify the operating logic, etc.

How would the coupling be implemented using the programmable logic below? What are the benefits of programmable logic?

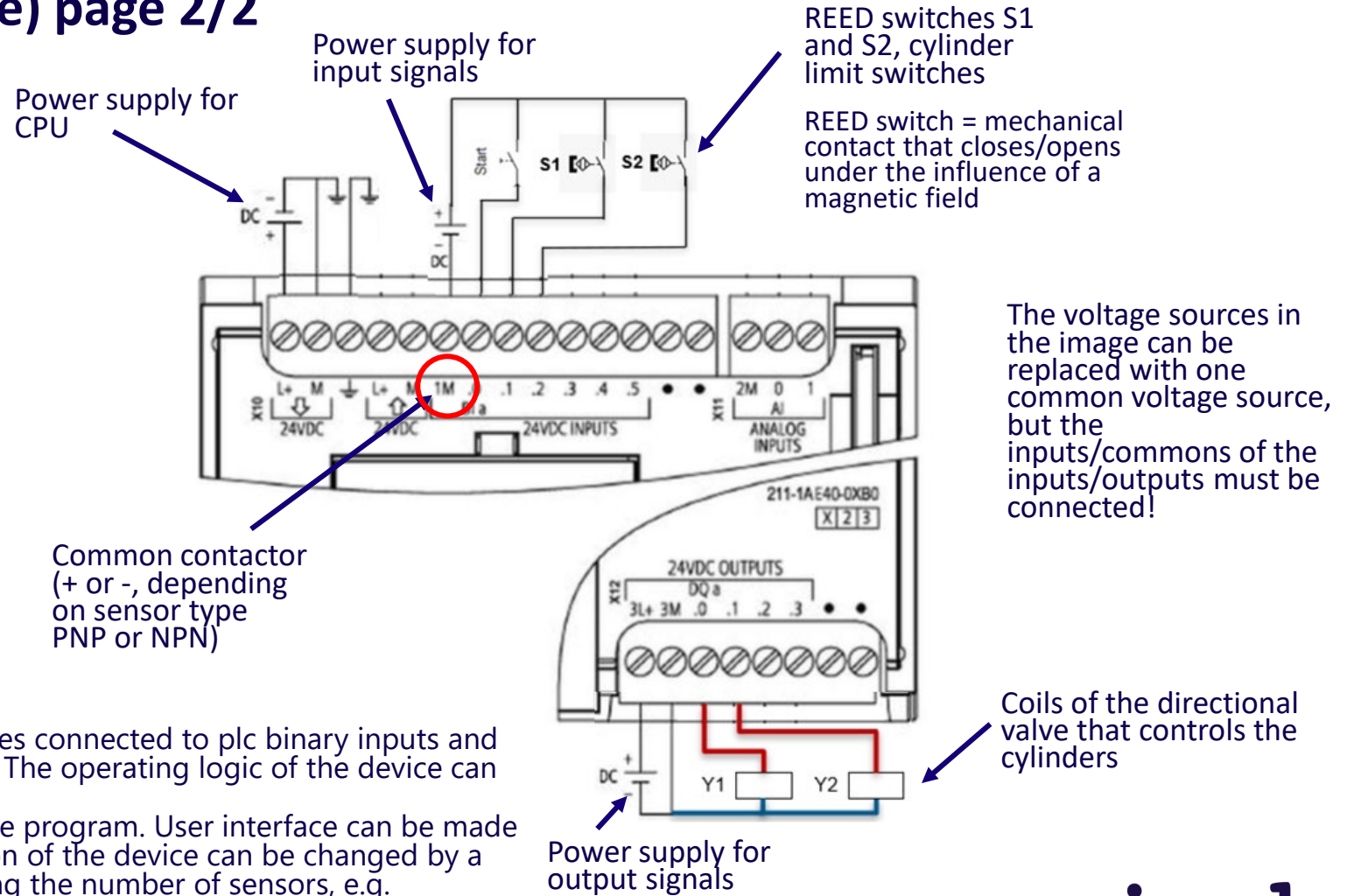


S7 1200 PLC
contactors

Example: changing the cylinder control to Siemens programmable logic (AutomationStudio example) page 2/2



After the change: Start button and limit switches connected to plc binary inputs and 5/2 directional valve coils to binary outputs → The operating logic of the device can be freely programmed using input signals! Counters, timers/delays, etc. are available in the program. User interface can be made for a touch screen or a computer. The operation of the device can be changed by a program change and, if necessary, by increasing the number of sensors, e.g. monitoring of the supply pressure by means of a pressure switch or transmitter and stopping the control in the absence of pressure or when it is too low.



Full connection examples of programmable logics, their modules, automation systems and other devices, and all other relevant and less relevant information can be found in the manuals (below is a screenshot of the attachment to the Siemens S7 1200 system manual)

